

TREATMENT PERFORMANCE SUMMARY

Alkalinity Enhancement in Biological Wastewater Treatment

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Executive Summary

This report summarises the outcomes of a laboratory-scale study evaluating the effect of alkalinity enhancement on biological wastewater treatment performance. Conducted using synthetic municipal wastewater in a sequencing batch reactor (SBR) configuration, the study compared a mineral-dosed reactor against an untreated control across multiple treatment cycles.

Key findings demonstrate measurable improvements in treatment kinetics, pH stability, sludge health, and effluent quality indicators, without adverse impact on parameters governing wastewater reuse.

Key Outcomes at a Glance

- Accelerated COD removal, especially in the first 1–2 hours of treatment
- pH stabilised above microbial optimum (≥ 6.5); control dropped to ~ 5.5
- TSS in treated effluent reduced by $\sim 50\%$ vs control
- MLVSS/MLSS ratio improved, indicating stronger microbial health
- Langelier Saturation Index (LSI) shifted from strongly corrosive to near-equilibrium
- No adverse impact on TDS or nutrient concentrations

1. Study Background & Objectives

Municipal wastewater treatment systems frequently operate under low or poorly buffered pH and variable organic loading, conditions that inhibit microbial activity, slow nitrification, and reduce overall treatment efficiency. Alkalinity supplementation addresses these challenges by stabilising pH and improving biological process performance.

This study was undertaken to quantify the treatment performance benefits of alkalinity enhancement in a controlled laboratory setting. The specific objectives were:

- Compare reactor performance under mineral-dosed and control (undosed) conditions
- Assess COD reduction rates and final effluent quality with and without alkalinity addition
- Evaluate the effect of alkalinity on pH regulation and its downstream impact on microbial activity
- Determine implications for tertiary treatment and wastewater reuse compliance

2. Experimental Overview

Two identical laboratory-scale batch reactors were operated in parallel using synthetic municipal wastewater prepared to OECD 303A specifications. Activated sludge was sourced from the Nerul

Sewage Treatment Plant (Navi Mumbai Municipal Corporation) and acclimated for three weeks prior to testing.

A four-hour SBR cycle was employed with 50% volumetric exchange per cycle (Fill → Aeration → Settle → Decant). A three-day baseline period was observed before any intervention, confirming comparable performance between reactors prior to the study phase. Alkalinity enhancement was then applied exclusively to Reactor A, while Reactor B served as the untreated control. Parameters monitored throughout included pH, dissolved oxygen (DO), COD, TSS, TDS, and MLSS/MLVSS.



Baseline reactor comparability confirmed:

Parameter	Reactor A (Pre-Intervention)	Reactor B (Control)
Initial COD (mg/L)	144–163	134–154
COD after 4h (mg/L)	57–76	57–76
DO post-aeration (mg/L)	6.5–8.3	6.5–8.1
Initial pH range	7.16–7.83	7.11–7.86

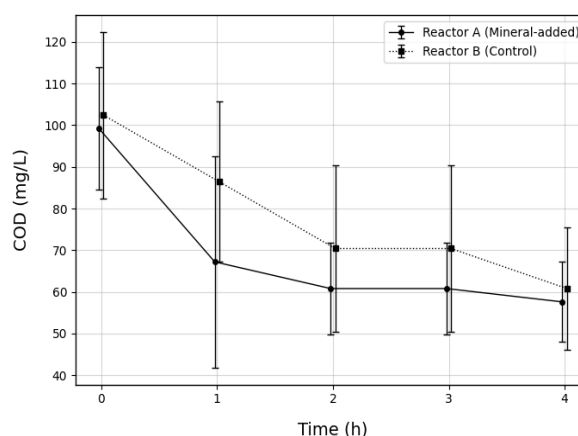
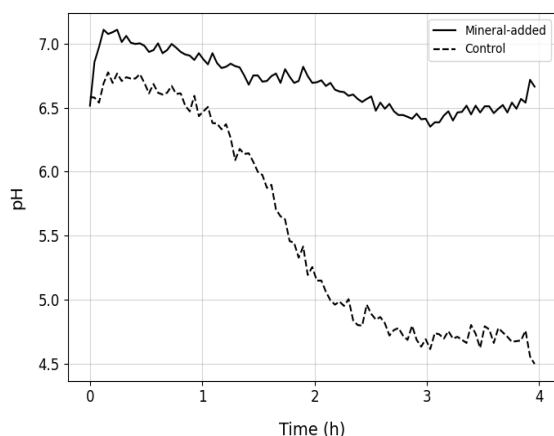
3. Treatment Performance Results

3.1 pH Stability

pH regulation was one of the most pronounced effects observed. In the control reactor (B), pH declined sharply during the aeration phase, dropping to approximately 4.5 by the end of the 4-hour cycle, well below the microbial optimum range of 6.5–8.5. This level of acidification is known to inhibit nitrification and destabilise the activated sludge process.

In the mineral-dosed reactor (A), pH was maintained at 6.69 at cycle end, within the MoEFCC discharge standard of pH 6.5–9.0. The alkalinity supplementation effectively buffered against biological acid production, sustaining conditions favourable for microbial activity throughout the cycle.

Time (h)	Reactor A, Mineral-Dosed	Reactor B, Control
0	6.52	6.58
4	6.69	4.50



3.2 Chemical Oxygen Demand (COD) Removal

COD reduction was faster in the mineral-dosed reactor, with the most significant divergence observed within the first two hours of aeration. By hour 1, the dosed reactor achieved ~67 mg/L versus ~86 mg/L in the control, a meaningful kinetic advantage. By hour 4, both reactors converged to comparable final COD values (~58–61 mg/L), indicating that alkalinity addition improves treatment rate rather than ultimate removal capacity.

The consistently lower starting COD in Reactor A across subsequent cycles (indicative of improved inter-cycle carryover treatment) suggests cumulative performance benefits with sustained alkalinity supplementation.

Time (h)	Reactor A COD (mg/L)	Reactor B COD (mg/L)
0	99.2 ± 14.7	102.4 ± 20.0
1	67.2 ± 25.4	86.4 ± 19.2
2	60.8 ± 11.1	70.4 ± 20.0
4	57.6 ± 9.6	60.8 ± 14.7

3.3 Total Suspended Solids (TSS)

TSS in the effluent of the mineral-dosed reactor was approximately 50% lower than the control at end-of-cycle, 45 ± 9 mg/L versus 91 ± 11 mg/L. The mineral-dosed reactor met the MoEFCC TSS discharge standard of <50 mg/L; the control reactor did not. This improvement is attributed to enhanced biological floc formation, consistent with the divalent cation-bridging mechanism documented in peer literature (Higgins et al., 2004).

TSS, Influent (mg/L)	Reactor A, Mineral-Dosed	Reactor B, Control
53 ± 1	45 ± 9	91 ± 11

3.4 Mixed Liquor Suspended Solids (MLSS) & Volatile Fraction (MLVSS)

MLSS and MLVSS were measured post-stabilisation to assess biological community health. The mineral-dosed reactor showed higher MLSS (3,160 vs 2,760 mg/L), within the optimal SBR operating range of 3,000 - 4,000 mg/L, and a significantly higher MLVSS fraction (2,568 vs 1,842 mg/L). The

elevated MLVSS/MLSS ratio in Reactor A indicates a more active and metabolically productive microbial community, consistent with the improved COD kinetics and TSS settling observed.

Parameter	Reactor A, Mineral-Dosed	Reactor B, Control
MLSS (mg/L)	3,160	2,760
MLVSS (mg/L)	2,568	1,842
MLVSS/MLSS ratio	0.81	0.67

3.5 Total Dissolved Solids (TDS)

TDS concentrations remained comparable between the two reactors throughout the treatment cycle and decreased slightly over time in both cases. Alkalinity supplementation did not measurably alter dissolved solids load - an important indicator for downstream reuse compatibility. This confirms that the intervention does not impose additional salinity burden on the effluent.

Time (h)	Reactor A, Mineral-Dosed	Reactor B, Control
0	196.7 ± 15.3	190.0 ± 10.0
4	176.7 ± 5.8	156.7 ± 11.5

4. Effluent Stability & Scaling/Corrosion Risk

4.1 Langelier Saturation Index (LSI)

The Langelier Saturation Index quantifies the tendency of water to deposit or dissolve calcium carbonate, a key indicator for infrastructure integrity and downstream reuse suitability. An LSI between -0.3 and +0.3 is considered operationally ideal.

In untreated effluent, LSI was measured at -2.0, indicating strongly corrosive conditions that can degrade pipework, membranes, and receiving infrastructure. Following alkalinity enhancement, LSI improved to -0.24, within the acceptable near-equilibrium range. This shift reduces corrosion risk without promoting scaling.

Parameter	Mineral-Dosed Effluent	Control Effluent
Langelier Saturation Index	-0.24 ✓ Within ideal range	-2.0 ✗ Corrosive

4.2 Tertiary Treatment Compatibility (Visual MINTEQ Modelling)

Geochemical modelling of secondary effluent chemistry confirmed that the added calcium species dissolved, over 93% of calcium as free Ca^{2+} ions and the majority of inorganic carbon as dissolved bicarbonate (HCO_3^-). Saturation indices indicated the effluent remained undersaturated with respect to most calcium carbonate mineral phases.

This indicates a low probability of uncontrolled precipitation during tertiary processes such as membrane filtration, disinfection, or polishing. The dominant effect of alkalinity addition is increased bicarbonate buffering capacity, which aids pH stabilisation in downstream treatment steps without imposing a scaling risk.

5. Wastewater Reuse Compatibility

All key parameters governing wastewater reuse suitability were assessed against CPCB and CPHEEO guidelines for irrigation, industrial cooling, and urban reuse applications. Mineral dosing had no adverse effects. The improved secondary effluent quality from the mineral-dosed reactor, specifically reduced TSS and stable pH, provides a superior feed to tertiary polishing stages, without affecting final reuse compliance or safety. Since tertiary treatment provides additional disinfection and polishing, maintaining comparable or improved secondary effluent quality indicates reuse standards would be met without modification to existing treatment trains.

6. Conclusions

This study demonstrates that alkalinity enhancement delivers measurable, multi-parameter improvements to biological wastewater treatment performance. The principal findings are:

- **pH Stability:** Alkalinity supplementation maintained reactor pH above the biological optimum throughout the treatment cycle, preventing the acidification (pH ~5.5) observed in the control. This protects microbial community function and sustains nitrification.
- **Accelerated COD Removal:** Organic matter degradation was significantly faster in the dosed reactor, particularly within the first 1–2 hours, offering operational benefits for shorter cycle configurations or higher-throughput applications.
- **Improved Effluent Quality:** TSS in the dosed reactor effluent (45 mg/L) met MoEFCC discharge standards; the control (91 mg/L) did not. Enhanced floc formation is the attributed mechanism.
- **Stronger Microbial Health:** Higher MLSS (within optimal range) and an elevated MLVSS/MLSS ratio confirm a more active biological community with better substrate degradation capacity.
- **Corrosion Risk Reduction:** LSI improved from -2.0 (corrosive) to -0.24 (near-equilibrium), reducing infrastructure corrosion risk without scaling risk.
- **Tertiary & Reuse Compatibility:** Geochemical modelling confirmed minimal scaling risk in tertiary treatment. Key reuse parameters (TDS, nutrients) were unaffected by the intervention.

Overall Assessment

Alkalinity enhancement is a low-cost, scalable operational strategy that improves treatment kinetics, effluent quality, and process stability in biological wastewater treatment systems. It is compatible with existing infrastructure, tertiary treatment processes, and established wastewater reuse standards, while providing a practical pathway for integrating carbon capture into treatment operations.